MUST Users Guide
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Chapter 1
General Description of MUST

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1.1. Overview

The capability to move power from one part of the transmission grid to another is a key commercial and technical concern in the electric utility environment. Transmission transfer capability is affected by load, generation, transaction patterns, and the need to be prepared for sudden equipment loss and emergencies. Engineers determine transmission transfer capability by simulating network conditions, including the effects of equipment outages and the effects of uncertainties; most significantly:

- Area-to-area transfers or transactions different than those assumed in the initial (base) calculation
- Generation dispatch patterns different from that assumed in the initial calculation and unavailability of specific generating units

The above uncertainties change the flow pattern of the initial (base case) FCITC calculation. Flow pattern changes can further load the limiting elements or cause other network elements to become limiting, thereby decreasing the transfer capability. The effect of uncertainties can be calculated by changing the initial calculation assumptions, rerunning the FCITC calculations, then merging and interpreting the results.

The purpose of the Power System Simulator for Managing and Utilizing System Transmission (MUST) is to efficiently calculate the impact of transactions on key network elements, identify the most limiting contingencies and constraints, calculate FCITC, and calculate FCITC sensitivity to transactions and generation dispatch changes. For example, the MUST functions for analysis of FCITC sensitivity to different uncertainties forms a quantitative basis for OASIS postings.

MUST efficiently calculates electric transmission transfer capabilities in grids the size of the U.S. Eastern Interconnection, and displays the impact of transactions and generation dispatch variations on transfer capabilities. The results are key to improving the confidence in posted transmission limits and managing the effect of grid transactions.

Several of the more important features include:

- Efficient calculation of fast interactive analysis of transfer capability for very large power systems
- Sensitivity analysis of incremental transfer capability with respect to other possible transactions or transaction schedules
- Text, tables, and graphical reports in the form of spreadsheets and charts
- A generation sensitivity analysis function which provides efficient tools for sensitivity analysis of import capability with respect to different generation patterns
- Efficient evaluation of the impact of any transaction on areas, zones, and subsystems, and user-defined monitored elements for both base case and various contingency conditions. Impact analysis results can be reported using graphical (bubble diagram) and ASCII output
- A built-in transaction scheduling system that can be easily interfaced to other transaction record-keeping files. Transaction scheduling can be used for adjusting the load flow case according to transaction schedules.
- A group of functions for fast contingency analysis (both AC and DC) providing interactive tools to examine and understand network conditions and limitations at a specified transfer level
- Distribution factors in a form useful to other applications such as Energy Management System (EMS) transaction curtailment procedures and OASIS posting

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- Versatile specification of input files, such as automatic subsystem definition and an EXCLUDE function that provides a convenient means to customize MUSTreports
- Efficient data structures and calculation methods tuned to the FCITC and FCITC sensitivity calculations

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1.2. Assumptions

MUST reports employs several load flow models including non-linear AC model and very fast linearized incremental model. The linearized model is subject to the following assumptions:

- For the linear transfer and DC contingency analysis functions, the assumptions inherent in the use of DC power flow for incremental flow changes are used
- The initial starting point for analysis can be obtained from the AC or DC load flow solution or from user-defined data (normally obtained from SCADA/EMS measurements)
- Contingencies are modeled using linear compensation methods.
- Subsystems (control areas) are defined using the area/zone definitions provided in the load flow base case. Bus-by-bus adjustments to the subsystem definitions is available.
- Transmission losses are allocated to individual buses across the system from an initial base case AC load flow solution
- Phase shifters can be set to operate either with fixed angles (locked) or with fixed power flows
- The MW flows of DC lines remain constant at the base case values. MUST can evaluate the impact of transactions within AC-connected areas. Transactions over DC lines are modeled by using intermediate buses.

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1.3. Linear Versus Non-Linear Power Flow Model

It is well know that the power flow model can be described by non-linear equations. Also, various transfer limit constraints, especially voltage stability related constraints, can be well described only by nonlinear models. At the same time, under most operating conditions, the level of non-linearity is generally weak, thus justifying the use of incremental linear power flow methods in the vicinity of the current operating point.

The question of which ATC power flow calculation model to use is widely debated. Probably, the best answer is a smart combination of linear and non-linear power flow models. MUST calculations are based on the incremental linear model around non-linear AC/DC/EMS starting flows. Users can verify linear results using the AC transfer limit verification and AC contingency analysis functions.

There are several reasons to help explain why linear methods still continue to be widely used for ATC calculation. Incremental linear power flow allows you to rapidly screen thousands of contingencies for the detailed power flow model covering very large interconnections such as the U.S. Eastern Interconnection with 30-40 thousand buses.

Probably, the primary reason for the active use of linear models is that the level of uncertainties intrinsic in ATC calculations far exceeds the errors due to linearization. These uncertainties being uncertainties in the future load levels and locations, network status, generators participating in the base dispatch, and generators involved in power transfers. Posted ATC values are always adjusted by several margins to reflect these uncertainties.

The lack of extensive data about external neighboring power provides another source of uncertainty. ATC calculations study the transfer limits between various control areas, while in most cases each control center has a detailed model of its own power system.

Another advantage of linear methods is the simplification of the results provided to other segments of the power industry such as marketers and end-point users. Transition from contract path to flow based reservation methods also represents an extremely important milestone. Flow based reservation methods rely heavily on the transfer distribution factors which are, under most conditions, fairly constant changing only with the topology of the power system. This is basically another way of describing the linear power flow model. These transfer distribution factors are readily available.

Rapidly developing power markets require very simple pricing signals assuming fixed MW pricing within reasonably wide ranges. Price variations due to non-linearity of the power flow model could represent an unjustified complication without any major advantages.

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1.4. Contingency Modeling in MUST

1.4.1. Modeling Contingencies

MUST uses exact AC solutions within the AC based contingency analysis and transfer analysis. Within the linear model, compensation algorithms are used to model both single- and multiple-event contingencies. When MUST starts, it computes the database of contingency compensation factors (CFAX) which are used by all other activities. CFAX factors are always recomputed whenever starting a case.

If a contingency creates an island, MUST automatically disables the buses in the island. By default, total power mismatch due to the loss of load and/or generation will be covered by the generation change at the correspondent swing bus. Such contingencies are classified as unbalanced, and users are strongly advised to review such contingencies for potential data errors. Branches or interfaces with at least one branch in the deactivated island will not be monitored.

MUST supports many of the features inherent in PSSE, although a few extensions that are included in MUST are not available in PSSE. There are only two limitations imposed when specifying contingencies:

- MUST does not allow contingencies that might connect AC buses which, in the base case, are AC connected to different swing buses. Also MUST does not handle load/generation transfers between such buses.
- MUST does not properly handle contingencies which result in swing bus separation. In this case, a new swing bus will not be selected for the system and will process correctly only buses that will be AC connected to that swing bus after the outage. Any group of buses not AC connected to the base case swing bus will be isolated.

Up to 100 elementary contingent events can be specified for one single contingency. Generator adjustments, as part of a contingency, are not included in the count for the 100 events per contingency limit.

Computing the CFAX database may take some time for large contingent and monitored files. In general, CFAX computation time is proportional to the total number of simple contingency events and is sensitive to the total number of monitored elements.

1.4.2. Phase Shifters Modeling

Overview of Phase Shifters Model

Modeling phase shifters during contingency analysis and transfer limit studies is often misunderstood. Selection of the proper phase-shifter model can have a significant impact on the results. Changing a phase-shifter model is likely to affect the transaction distribution factors in the neighborhood of regulating phase shifters causing varying ATC/TTC results in that area. In general, these issues should be reviewed with respect to all elements of the power system, such as FACTs devices, that can efficiently control MW flows without changing generation or load.

In general, depending on the objective of the study, you may want phase shifters modeled in one of the following ways:

- Disable the phase shifters altogether, resulting in fixed phase angles throughout
- Allow the phase shifters to control flows during the base case run but then disable the phase shifters (fixed angle) during the contingency analysis. This configuration is quite common.

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- Follow the practice of some, primarily metropolitan areas, in which phase shifters control flow on the underground cables. In this case, utilities allow phase shifters to be adjusted according to the desired flow settings under contingencies as well as in the base case. The rationale behind this choice is that if post-contingent branch flows are below short term emergency ratings (STE), then phase shifters may help to reduce flows on overloaded branches below long term emergency rating (LTE) within the allowable 10-15 minutes time frame for STE.

- In some planning applications a user may be interested in determining a set of centralized corrective phase shifters adjustments that will help reduce post-contingent overloads through centralized phase-shifter redispatch. Under this scenario modeling of phase shifters is complicated because some phase shifters may control flows according to their local objective of keeping through-flows within designated ranges, while others are assumed to be globally optimized.

- In some applications, users may choose to use phase shifters in a preventive mode. Here you are interested in determining an optimal phase-shifter redispatch to be used in the base case that would reduce or eliminate various contingent overloads.

- Some phase shifters can be controlled to maintain flow on the selected branches/flowgates within desired limits and are adjusted only when this limit is reached.

MUST allows three options for phase-shifter modeling in the Linear DC Model:

- Fixed

- Constant Flow in Base Case, Fixed Angle in Contingencies

- Constant Flow for Base Case and Contingencies

In the AC Model, an option can be set to allow Phase Shift Adjustments. When checked, this option is equivalent to the Constant Flow for Base Case and Contingencies described above. Otherwise, the default is the Fixed Angle option.

The Linear DC load flow model does not observe the regulation of phase-shifters angle limits, (i.e., angle regulators have unlimited capabilities). This simplification results in transaction distribution factors (TDF) which do not change with transfer level. The AC load flow model, with the Phase Shift Adjustments selected, however, always observe phase-shifter limits and may result in a significant differences in TDF factors for different transfer levels. This can essentially complicate the analysis of transaction interaction on the ATC/TTC results.

The recommended solution method for AC based contingency and transfer limit calculations is the Fast Decoupled load flow solution method with Phase Shift Adjustments selected.

Phase Shifters Controlling Flows in the Base Case, But Disabled During Contingencies

The most commonly used phase shifter model is to assume that the phase shifters are controlling the flows but are fixed during a contingency. This assumption is based on the NERC N-1 criteria that phase shifters are not fast enough to control flow immediately after a contingency. This difference in phase shifter modeling from the base case to the contingency case brings an additional complexity that may not be intuitive. Though phase shifters are modeled as being fixed during the contingency analysis, phase shifters are, in fact, locked at the prescribed transfer level when the contingency happens. As a result, phase shifter angles are dependent on the study transfer level. In other words, even though you may have selected that the phase shifter angles be fixed under contingencies, they do change with study transfer level as necessary to accommodate the base case.

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The above discussion, of course, only pertains to the linear DC model as the AC calculation only allows modeling phase shifters as either locked or controlling flows under contingencies as described in the previous section. One possible approach around this limitation is to solve each base case AC load flow to get the initial phase shifter settings then solving the contingency AC load flows with phase shifters locked.

As phase shifters are often modeled with fixed angles in contingencies, it is tempting to assume that the OTDF (contingency) factors are always the same, irrespective of the phase shifter modeling option used for the base case. This is not the case, however, as described below:

For single branch contingencies, OTDF factors for any branch are computed using the following formula:

OTDF = PTDF + LODF * cPTDF (1.1)

where:

Table 1.1. Calculated Factors

OTDF	Outage transfer distribution factor (impact of transaction under a contingency)
PTDF	Base case power transfer distribution factor, with phase shifters either controlling flow or have fixed angles
cPTDF	Base case power transfer distribution factor on a contingent branch with phase shifters either controlling flow or have fixed angles
LODF	Line outage distribution factor on a contingent branch with a monitored element assuming phase shifter angles are fixed

Assume that regulating phase shifters cannot become overloaded in the base case because they have fixed through-flows (in the DC model) and that the base case response factors (PTDF) for regulating phase shifters are always zero for all possible transactions. If the regulating phase shifters can be impacted by transactions under contingency conditions, then, under a contingency, regulating phase shifters can become overloaded because phase shifter flow can change. Though phase shifters are always modeled with fixed angles in contingencies, the OTDF factors are computed assuming fixed angles in the base case, which are different from OTDF factors computed when phase shifters are controlling flows in the base case.

PTDF and cPTDF used in the formula above are sensitive to the way phase shifters are modeled so, for example, if LODF is zero (i.e., contingency has no impact on the monitored element), then OTDF is equal to PTDF value.

Changing the MW set point for regulating phase shifters or changing the angle for fixed phase shifters have similar effects on the remaining power system as providing an additional transaction between phase shifter terminal buses.

Contingencies With Dynamic Bus Events

When there is an interaction between study transfers and contingency specifications, the amount of power injections due to contingencies cannot be determined using the initial load flow case conditions since it changes with the study transfer level. Examples of these types of contingencies are:

- A contingency involving a generator which also participates in the study transfer where the amount of MW generation loss for that contingency depends on the transfer level
- Loads lost in an island or moved to other buses as the result of a contingency where the importing system was defined to model load scaling involving buses in the island. Here the load loss/transfer depends on the transfer level

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- Maximum output from a plant is limited by the loss of that plant itself

MUST and PSSE differ in how they handle these contingencies with dynamic bus events. MUST models all contingency injections using the current transfer level, while PSSE uses the initial level of generation or load. As a result, MUST users may find some differences in FCITC limits reported for contingencies such as those described above.

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1.5. MUST Limiting Elements

1.5.1. Branches, Interfaces and Flowgates

Three types of the constraints are supported by MUST:

- Monitored branches
- Monitored interfaces, where the flow on an interface is taken as the sum of the flows on the branches comprising the interface. Interfaces quite frequently are used as proxies to represent voltage and stability limits and sometimes commercial limits on interchanges between control areas.
- Monitored flowgates. Flowgates are designated as a subset of proxies for transmission limitation and transmission service usage. The concept of flowgates has been actively developed under NERC initiatives for sharing transmission constraints for TLR (transmission loading relief).

MUST uses two sets of ratings for every monitored branch and interface: base case and contingent. By default, contingency violations are checked (using the same contingent rating) for all specified contingencies. The EXCLUDE function allows the user to disable checking violations for designated contingencies.

Flowgates are defined together with other monitored elements (branches, interfaces) within the monitored element definition file. All functions including FCITC limits analysis, parallel transfer impact analysis, generation sensitivity flowgate constraints as well as other types of constraints.

The major difference between branches/interfaces and flowgates is that, while monitored branches and interfaces are normally checked in the base case and all contingencies, flowgates are checked only in the base case (for the base case flowgates) or for only one contingency, as specified in the definition of the flowgate.

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Chapter 2
Computing Transfer Limits

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2.1. Overview of the FCITC Calculations in MUST

FCITC measures the maximum increase in power transfer that can take place between two selected subsystems without violating branch ratings or interface limits under a single contingency. Traditionally, this analysis has been performed using linearized models with generation in the sending and receiving subsystems adjusted according to user-defined participation factors.

MUST offers two functions for analyzing transfer limits:

- **Linear FCITC Calculation - Single Transfer** Use this function to calculate the thermal transfer limit, and identify the elements causing the limit between a defined sending and receiving subsystem.
- **Linear FCITC Calculation - Multiple Transfers** This function reports transfer limits for multiply-selected sending and receiving subsystems. Using subsystems defined in the input files, the user may select any pairs of subsystems as the sending and receiving subsystems for this analysis.

The above two functions are also available for calculating transfer limits using full AC power flow techniques:

- **AC Based FCITC Calculation - Single Transfer**
- **AC Based FCITC Calculation - Multiple Transfers**

The AC transfer limit calculation involves the AC verification of a defined number of the most limiting linear calculation limits.

MUST allows overlapping (in fact, arbitrary) subsystems to be used for FCITC definitions for transaction analysis. This allows the user, for example, to model transfers from generation to load within the same area to simulate daily pattern and analyze loop flows. The same generators could be defined in both sending and receiving subsystems, within parallel transfer sensitivity analysis.

Fcitc dialogs have a checkbox named "show old report". If this option is checked, the report will be in the text format. Otherwise, the reports will be in table format.

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2.2. Linear Transfer (DC Based) Limit Calculations

2.2.1. Detailed FCITC Report

Creating FCITC Detailed Report

Transfer Limit Analysis > Linear FCITC Analysis > Single Transfer

Figure 2.1. Linear Transfer (DC Based) Dialog

A FCITC detailed report is accessed by selecting the Single Transfer button under the Linear FCITC Calculation group from the Transfer Limit Analysis Menu Item. Here the user may specify the study transfer, by choosing previously-defined subsystems as the desired FROM and TO participants and entering the maximum incremental transfer in MW to be studied.

There are three formats available for the basic FCITC report:

- Two-line
- Wide
- TLTG

There are different ways to present the incremental transfer capability for the study transfer:

- A report on the incremental transfer level of the study transfer (so-called FCITC report) showing incremental transfer levels at which limiting monitored elements reach their thermal limits under the identified contingency. Check the FCITC violations box to select this reporting option. This is the default format and is the one which will be presented if no report boxes are checked.
- Study transfer capability in terms of the interface flow instead of the incremental study transfer level. In this case, the report includes the interface flow levels at which the reported limiting monitored elements/contingencies reach their limits. Check the Interface Flow box to select this reporting option and choose the interfaces to report.
- To present the transfer capability information to show the total export level for the selling subsystem at which the limits occur, or the total imports for the receiving subsystemc check the Export level and/or Import level boxes to select the desired reporting option(s).
- To see flows on several flowgates at the identified maximum transfer level check the Flowgate Flow box and choose the desired flowgates.

The user also has the option of altering the basic report by adding a separate base case report, by including a detailed report on generation adjustments required to increase the transfer from the base case to a separately-specified maximum incremental transfer that is input by the user, and by suppressing the reporting of existing overloads that would be relieved by the studied transfer. These options are independent of each other and can be selected by checking the respective boxes at the bottom of the Linear Single Transfer Limit dialog.

Reading FCITC Reports

MUST can create an FCITC report in three different formats:

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- A detailed Two-line report, which is also the most detailed FCITC report, allowing convenient analysis of all relevant information on one screen.
- A Wide report uses one line per every reported limiting element. This report may be easier to edit than the Two-line report when embedding it into user's documents. This report contains most of the same information as the two-line report, omitting the PTDF and the precontingent flows on the contingency branch.
- A report in the same format as PSS®E activity TLTG, which may be useful for users who already have automated procedures that assume the layout of TLTG reports. This format does not contain as much information as the other two formats.

All MUST FCITC reports contain an ordered list of limits for monitored element/contingency pairs. The FCITC reports are sorted such that the smallest (i.e., the most restrictive) limits appear first. The maximum FCITC of interest is user specified. Study transfer levels at which branch and interface limits are reached for the base case conditions are also listed within the FCITC report.

A separate base case only violations report (NITC report) can also be included within the FCITC report.

The fourth column "N" in the Two-line FCITC report is a violation counter. An "NA" in this column denotes a situation in which a base case violation is improved by the transfer, and an "R" will be shown on the line as well. Due to skipping the reporting of violations for the same limiting element above the maximum specified, there may be a skip in the numbers under "N".

The next column (with the header FCITC) for all report formats always contains the incremental transfer level between the two study systems where a monitored element or interface reaches its limit for a specified contingency. The FCITC reports are sorted by ascending FCITC values. Transfers lower than reported in the first-row transfer level (N equals 1) would not violate the rating of any branch specified as a monitored element or the limit of any specified interface or flowgate under either base case conditions or under any of the contingencies that were studied.

The next column designates the limiting monitored element (L:) and limiting contingency (C:). The same monitored element may be reported several times under different contingencies. The number of times that a particular monitored element will be listed in this report is limited to the maximum number specified in the MUST Options. If this limit is reached, the last entry will be marked with an asterisk (*). Base case limits are always reported even if the number of entries for a particular monitored element is exceeded, unless the monitored element is specifically excluded by the user. An "R" appearing before the "L:" denotes a violation removed by a transfer at or above the indicated FCITC value.

The Ncon column provides a contingency index that can be used to identify this contingency within the various MUST activities and reports.

The Pre-Shift Flow column contains the postcontingency flows on the monitored elements under the specified contingency at the base or initial transfer level, in other words, zero study transfer level.

The sign in the various MW Limit columns indicates the direction of flow on the limiting element. A positive value indicates flow from the FROM bus to the TO bus for limiting branches. For interfaces, a positive limit indicates the total positive flow from interface FROM buses. A negative sign (-) indicates the opposite flow direction.

The TDF column contains OTDF for the contingency-limiting constraints with the outage of the indicated contingency. For base case limitations, the TDF column indicates the PTDF (the base case distribution factors) of the current study transfer on the reported limiting element.

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The LODF column shows LODF of a single branch contingency on the monitored element.

PTDF column contains the distribution factor for the study transfer on the reported limiting element or on the contingency branch.

The remaining two columns provide the pre-contingency flow on the limiting element and contingency branch at base study transfer level and at the transfer level equal to the FCITC value (i.e., when the monitored element or the interface reaches its limits for the reported contingency).

Understanding Initial Violations

Under certain conditions, the FCITC report might contain one or more negative FCITC entries. This happens only if the original load flow case (before the study transfer) contains contingency violations on monitored elements that have been specified for this study. The reported FCITC value is equal to the incremental reduction in transfer level where a particular monitored element/contingency pair is no longer limiting. It is recommended that the user carefully review any reported negative FCITC values, as some of the MUST activities (e.g., parallel transactions sensitivity analysis) cannot proceed when there are rating or limit violations for the base transfer case. The EXCLUDE function allows exclusion of monitored elements, contingencies, or monitored element/contingency pairs that may cause negative FCITC values. Using the EXCLUDE function also allows the user to correct erroneous ratings without updating the load flow case.

If the original load flow case (e.g., at zero study transfer level) contains violations, it may significantly complicate FCITC analysis reports. All initial violations can be classified into three categories based on the study transfer impact:

1. The study transfer has a response factor below a user-specified threshold. In this case, initial violations will not be reported at all. Quite often, generation sensitivity analysis functions may report such constraints as the limiting one for one of the worst dispatch patterns.

2. Increasing the study transfer level makes the violation even worse. In this case, MUST reports FCITC for that constraint as negative, which means that reverting the study transfer to the reported negative transfer level will eliminate this violation.

3. MUST will report initial violations that will be removed by an increase in the study transfer simultaneously with FCITC violations. This feature helps the user to assess the feasibility of the study transfer. For example, if all initial violations will not be removed before a study transfer increase causes new violations, then there is no feasible level of study transfer without at least one violation. Violations removed by the increase in the study transfer will be indicated by an "R" in the same column where MUST indicates violations that were reported the maximum number of times as specified in user options. An "R" can be found on both single FCITC and Multiple FCITC reports.

2.2.2. Multiple Transfer FCITC Report

Creating the Multiple Transfer FCITC Report

Transfer Limit Analysis > Linear FCITC Analysis > Multiple Transfers

Figure 2.2. Multiple Transfer (DC Based) Dialog

The Multiple Transfer FCITC Report is accessed by selecting Multiple Transfers under the Linear FCITC Calculation group from the Transfer Limit Analysis Menu Item. By selecting this option, the FCITC Study Multiple

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Transfer Selector dialog is displayed. MUST calculates the FCITC limits simultaneously for many transfers, and then presents the results in a summary table. Each transfer can have a different maximum transfer level.

To change the Transfer limit levels of some or all defined transfers, enter a new test level in the Transfer Limit box, select the transfer(s) you wish to change, and click the **Modify** button. To delete selected transfers, select the transfer(s) you wish to delete and click the **Delete Transfer** button. To clear the entire list of transfers, click the **Clear List** button.

A text box at the bottom of the dialog labeled Maximum Number of Violations allows you to specify the maximum number of violations to report per transfer, regardless of the total number of these violations.

The check box labeled Show violations removed by transfer level increase allows the user to choose whether or not to show those initial violations that are removed by increases in the transfer level.

The Multiple Transfer FCITC report can be presented in one of five available report formats:

- Two-line
- Wide
- TLTG
- Data processing two line
- Data processing wide (Suppress events)

The data processing formats provide results with data repeated as necessary so the final results can be possibly exported.

Also when requesting a "Data Processing Wide" format report you can request that multiple events for each contingency be suppressed by checking the Suppress Events checkbox. This will result in each limit taking up only 1 row, creating a more compact report.

There is a check box available for the user to specify the type of report desired as Base case only. If the Base case only box is checked, the Multiple Transfer FCITC Report will contain only base case limit information, i.e., NITC.

The Multiple Transfer FCITC Report has a similar structure to the Single Transfer FCITC Report. The explanation of the data fields and column headers is consistent with the Single Transfer FCITC Report. An "R" will appear in the FLAG column to indicate a pre-transfer overload that is removed by the given transfer.

If a study transfer was defined prior to defining the transfers for the FCITC Summary Report, then that study transfer will be restored upon completion of the summary report. If a study transfer was not defined, the last transfer appearing in the Study Transfer box will be designated as the study transfer.

FCITC Analysis in Command Line Mode

To obtain FCITC and NITC limit reports, the user must first define study transactions, using subfunction STUDY under function TRANSFER. When selected, subfunction STUDY requires specification of sending and receiving subsystems and the maximum transfer level in MW. Subsystems can be specified either by exact subsystem names or by subsystem numbers that can be obtained using subfunction SYSTEM under function TRANSFER. After defining a study transfer, MUST reports the total number of calculated limits below the maximum transfer level specified.

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An FCITC two-line detailed report can be obtained using subfunction VIOL. Function BVIOL reports base case limits only (NITC report). The wide report can be obtained by using the WVIOL function. Function TLTG will produce a report in TLTG format. All of these functions will accept a numeric parameter specifying the maximum number of violations to report. The FCITC analysis in the command line mode is structured to perform a single transfer at a time.

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2.3. AC Transfer Limit Verification

The objective of the AC Transfer Limit Verification function is to find true AC limits, based on thermal constraints only. This function uses the results of the DC transfer limit analysis to screen for the most limiting constraint/contingency pairs to study in AC mode. Rather than run full AC transfer limit calculations it solves only a few AC load flows to verify the first few violations identified by DC linear methods.

The execution of the AC Transfer Limit function in MUST can be described as follows:

- MUST runs DC based transfer limits analysis and identifies a number of the monitored element/contingency pairs which result in the lowest FCITC values for the study transfer.
- MUST then runs AC parametric transfer analysis (several AC load flows) to maximize the study transfer level until the limit for the appropriate limiting constraint/contingency pairs is reached.
- If the AC Load Flow diverges, it will find the maximum transfer level based on load flow solution convergence.

MUST supports two AC-based FCITC calculation functions: Single Transfer and Multiple Transfers. These are accessible from the AC Based FCITC Calculation group from the Transfer Limit Analysis Menu Item.

Selecting Single Transfer from the AC Based FCITC Calculation options displays the AC Single Transfer Limit dialog.

The AC Single Transfer Limit dialog allows the user to specify the transfer and the test level to be studied. The AC transfer limit tolerance may be modified. The user specifies the number of First DC (Linear FCITC) violations to examine in the AC solution process. MUST will first identify the most limiting linear transfer limits and then test these using a full AC power flow solution.

Options for the desired AC Load Flow Solution Method and for handling a Non-Divergent Load Flow are provided.

2.3.1. AC Transfer Limit Report

Transfer Limit Analysis > AC Based FCITC Analysis > Single Transfers

Figure 2.3. Single Transfer (AC Based) Dialog

The AC Transfer Limit Report contains the AC and DC limits for the specified transfer. The delta between AC and DC is included.

AC Transfer Limit Report Codes

The following are definitions of comments that appear on the AC/FCITC report. These definitions provide more information about the comments on the AC FCITC report for situations in which there is a violation prior to the transfer.

- "Removed" is the same as "R" on a Wide FCITC report and means that the transfer has "Removed" the violation at the value under the AC/FCITC.

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- "NoLimit" indicates that either that you cannot correct a violation that existed prior to the transfer, or, that there is no longer a problem by the level of the transfer, such as was found in the linear (DC) case. "NoLimit" always shows the FCITC at the transfer level and the post shift calculated there. When "NoLimit" is reported, it means that MUST tested the AC LF at the max test level, it converged, but on the way to the max test level the limit was never crossed.
- "LowSens" (low sensitivity) is an AC_TDF which has fallen below cutoff.
- "TDF_sign" indicates there has been a change in sign in the TDF from DC to AC, so the DC screening was useless.
- "NotConv" solution did not converge. An increase in the transfer causes the LF to blow up before reaching the expected transfer limit. The maximum transfer level based on solution convergence is shown.
- "NotConv-" solution did not converge. A decrease in the transfer causes the LF to blow up before reaching the expected transfer limit. The minimum transfer level based on solution convergence is shown.
- "No_Sens" solution did not converge. Indicates AC_TDF is so low that it has no effect (no sensitivity).
- OK means that a solution was found and the algorithm converged.
- ##Excluded means that the contingency was excluded.
- ##Disabled means that the contingency was disabled.
- NotC_ND means that the algorithm did not converge using the non-divergent solution method.
- OK_NonD means that the algorithm converged using the non-divergent solution method.

2.3.2. AC Multiple Transfer FCITC Calculations

Transfer Limit Analysis > AC Based FCITC Analysis > Multiple Transfers

Figure 2.4. Multiple Transfer (AC Based) Dialog

This function has a similar look and feel to the Linear Multiple Transfer function. The user may choose any combination of sending and receiving systems and may choose the test transfer level for each sending/receiving pair. The user may also import/export a transaction/description file.

The function provides the same AC LF Solution Options described above, except that only Standard and Data Processing report formats are available. The data processing format provides results with data repeated as necessary (i.e., no empty cells) so the final results can be sorted and exported. Also, with the data processing format, you can request that multiple events for each contingency be suppressed by checking the **Suppress Events** checkbox. This will result in each limit taking up only 1 row creating a more compact report.

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Chapter 3
Transfer Limits Sensitivity Analysis

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3.1. Analysis of Transactions Impact on Transfer Limits

3.1.1. Parallel Transfer Detailed Report

Transfer Limit Analysis > Transfer Sensitivity Analysis > Parallel Transfer Detail

Figure 3.1. Parallel Transfer Dialog

The Parallel Transfer Detail function provides an analysis of the impact on the specified study transfer FCITC for a change in a single parallel transfer. This report is accessed by selecting Parallel Transfer Detail under Transfer Sensitivity Analysis group from the Transfer Limit Analysis Menu Item. The user has the option of defining the study transfer or retaining the existing study transfer (if one were defined) and defining the parallel transfer. Three types of reports are available: Summary, Detailed report or both.

Before performing a transfer sensitivity study, you must specify a study transfer and compute the study transfer FCITC determined by the most restrictive constraint. When a change in the parallel transfer changes the flow on the most restrictive monitored element (branch or interface), the study transfer FCITC limits will also change. If a parallel transfer decreases the flow on the limiting element, the FCITC limits will increase. If a parallel transfer increases the flow on the limiting element, the FCITC limits will decrease.

It is possible to use study and parallel transfer network sensitivities (OTDF factors) to derive the Parallel Transfer Response Factor (PTRF). This factor is the ratio of change in study transfer FCITC to the change in parallel transfer. The PTRF can be computed by using the study and parallel transfer network sensitivities (OTDF factors) as shown below:

```
PTRF = (d)FCITC / (d)ParDisp = -OTDFpar / OTDFstudy

       where:

(d)ParDisp =  Change in the parallel transfer level.
(d)FCITC   =  FCITC change due to additional parallel transfer.
OTDFpar    =  OTDF factor for the parallel transfer on the limiting element.
OTDFstudy  =  OTDF factor for the study transfer.
```

PTRF defines the slope of a linear segment whereby if PTRF is positive, an increase in a parallel transfer results in an increasing FCITC for the study transfer. At some transfer level, another constraint may become more restrictive than the initial one resulting in a change in slope or a PTRF change.

As a result of using a linearized load flow model, all transfer sensitivity reports describe a piecewise linear concave function. It is concave because every new more restrictive segment has a smaller or increasingly negative PTRF value.

Transfer sensitivity analysis provides answers to the following questions:

- What constraints are the most limiting at the different parallel transfer levels?
- At what parallel transfer level does a new constraint become the most restrictive?
- What are the PTRF values for the most restrictive constraint?

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In most cases, there are not more than two or three most restrictive constraints within a relatively large range of change in parallel transfer. If there is only one limiting constraint, a user can use the linear extrapolation to identify FCITC at different parallel transfers levels using the PTRF value.

The transfer sensitivity analysis also provides reports on flows and PTDF, LODF, and OTDF factors for the most restrictive constraints.

All these reports show how the study transfer FCITC changes with parallel transfer changes. At every parallel transfer level, a study transfer FCITC is determined by the most restrictive contingency/monitored element pair for the selected study transfer. Thus, to create a transfer sensitivity report with respect to the second limiting constraint, the user must exclude the first most limiting constraint using the EXCLUDE feature.

Under some conditions, the FCITC report might contain negative FCITC values. MUST will not generate summary and detailed reports for this condition.

3.1.2. Multiple Study Transfers

Transfer Limit Analysis > Transfer Sensitivity Analysis > Multiple Study Transfers

Figure 3.2. Multiple Study Transfer Dialog

Multiple study transfer reports provide a convenient way to identify how one additional transaction can impact a whole group of study transfers. These reports are useful for operational applications (OASIS posting) where a transmission provider wants to adjust the posted interface limits based on an additional transmission service reservation or schedule.

This report is accessed by selecting Multiple Study Transfers under Transfer Sensitivity Analysis group from the Transfer Limit Analysis Menu Item. Press the "Select" button on the Multiple Study Transfers dialog to select study transfers to be specified against one parallel transfer to be included in the report.

A summary table provides a simple and convenient way of updating FCITC values in the operation environment. This type of table can be created for every single parallel transfer of interest and should be recomputed when schedule changes if there are study transfers with multisegment FCITC curves.

3.1.3. Multiple Parallel Transfers

Transfer Limit Analysis > Transfer Sensitivity Analysis > Multiple Parallel Transfer

Figure 3.3. Multiple Parallel Transfer Dialog

The multiple parallel transfer sensitivity report displays the impact of a selected set of parallel transfers on one study transfer FCITC.

This report is accessed by selecting Multiple Parallel Transfers under Transfer Sensitivity Analysis group from the Transfer Limit Analysis Menu Item. Press the "Select" button on the Multiple Parallel Transfers dialog to select the study transfer as well as the parallel transfers to be included in the report.

In general, the simultaneous impact of two or more parallel transfers on the study transfer FCITC cannot be obtained directly. The constraining contingency/limiting element pairs may be for different parallel transfers.

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In that case, the impacts of individual parallel transfers on FCITC limits cannot be added to obtain the total simultaneous impact on the study transfer. However, such superposition is possible to the left of the first inflection points. For this range of parallel transfers, the constraining contingency/monitored element pair is the same pair that determines the nominal study transfer FCITC when all parallel transfers are zero.

Multiple parallel transfer reports on selected study transfers can be very useful for identifying potential parallel transactions that may have a significant impact on the selected study transfer.

If a study transfer was previously defined, it will be restored after producing the requested reports. However, if a study transfer was not defined, this function sets the study transfer as the last transaction shown in the Define Study Transfers scroll box.

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3.2. Generator Sensitivities Analysis

3.2.1. Applications of Generator Sensitivity Analysis

Most FCITC computations are based on the assumption that generation participation factors for sending and receiving subsystems are user defined and do not change. In this subsection, the abbreviation FPF FCITC is used to denote FCITC analysis computed using Fixed Participation Factors (FPF). This is the simplest method of FCITC analysis. However, under many conditions generator participation factors are unknown or the user may have an option to select participation factors and generators to be used in a transfer.

In using FPF, the user should be aware of the following questions and considerations:

- How will study transfer FCITC change if a user specifies different participation factors?
- How will study transfer FCITC change if a user specifies different generators to participate?
- Which are the most limiting contingencies/monitored elements that limit study transfer with respect to all possible combinations of participating generators and participation factors? Did the FCITC analysis actually find the most limiting contingency?
- What are the possible FCITC variations due to different dispatch patterns? The answer to this question may have an impact on the determination of the transmission reliability margin (TRM).
- Which are the generators in the importing systems that have the greatest positive impact on FCITC? These generators can be defined as the must-run units for a selected study transfer. Generators in the importing system that must run with respect to every import scenario can be defined as the must-run units for the control area.

All of the above questions are difficult to answer using traditionally available tools. Multiple FPF FCITC runs could help answer some of these questions, but cannot provide a complete solution considering all the possible dispatch variations.

FPF FCITC calculations do not check generator limits and do not allow study transfer specification at any transfer level. It is the user's responsibility to check that all generators will be within these limits for the selected transfer level. This approach simplifies subsystem definitions and data maintenance. It is, for example, common to specify only a few representative buses as generation sources and sinks. However, this approach may result in violation of generator capacity limits and associated incorrect FCITC limits.

The Generation Sensitivity Analysis (GSA) was designed to answer these questions. Within GSA, MUST always observes generator limits. For this reason, if an FPF FCITC analysis determines limits that do not appear in the GSA calculation, the FPF FCITC calculations will result in artificial limit violations caused by generation dispatch violating generator limits.

If we define a study system as that subsystem for which the user is requesting an FCITC sensitivity due to variations in participation factors, the GSA function finds the dispatches in the study system (importing or exporting) that minimize study transfer FCITC (i.e., the worst generation dispatch). The GSA function can determine:

- Generators in the importing/exporting systems whose output adjustments (decrease or increase, respectively) would result in the lowest import levels.
- ALL contingencies and constraints that may be limiting with respect to all possible combinations of generator adjustments in the importing/exporting systems.

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- Transfer security margins. If the worst dispatch pattern cannot create violations by adjusting user-specified generators, then the specified transfer level is secure with respect to any dispatch in the study system.

In most cases, the GSA function determines several of the worst dispatches for different monitored elements/contingency pairs. As a result, the worst dispatch analysis may find more limiting contingencies/constraints than the FPF FCITC analysis. GSA analysis can also point out some local problems that are not related to the study transfer FCITC. Several effective ways to use GSA are to:

- Include all generators in the importing or exporting system.
- Examine the contingencies/monitored element pairs to identify local problems and must-run generators.
- Remove the local must-run generator from the importing subsystem list, customize monitored elements using the EXCLUDE file, and repeat analysis.

Currently, MUST assumes that the opposing system is adjusted according to fixed participation factors. Therefore, for a specific transfer level, reduction/increase in the dispatch of a generator in the study subsystem is made up of fixed participation factor adjustments for the opposing system.

It is important to understand the difference between the worst generation dispatch method and the optimal power flow (OPF) based approaches method. These two methods have opposite objectives. When OPF is used for transfer studies, it is used mostly to maximize transfer capability. In contrast, the GSA function finds the worst dispatch pattern that creates violations, e.g., minimizes transfer limits. If the GSA function cannot create violations, then this proves that the specified transfer level is secure. Therefore, OPF and GSA functions find the upper and lower boundary values for transfer levels between selected subsystems.

3.2.2. Generation Sensitivity

Transfer Limit Analysis > Transfer Sensitivity Analysis > Generation Sensitivtiy

Figure 3.4. Generation Sensitivtiy Dialog

The GSA function requires almost the same input data as an FPF FCITC analysis (exporting/ importing subsystems and desired transfer level). The only difference is that a user should modify the definition of the importing subsystem to add all generators to be adjusted when determining the worst dispatch pattern. GSA reports compare FPF and GSA violations. The user must specify nonzero participation factors in the subsystem definition file for generators to be used by the FPF FCITC analysis, and zero participation factors for generators to be used by the GSA function only.

The maximum number of worst case scenarios to report is 500,000.

This report is accessed by selecting Generation Sensitivity under Transfer Sensitivity Analysis group from the Transfer Limit Analysis Menu Item. Then enter the study transfer, review the relevant MUST options, and select the type of report. This function redefines the current study transfer.

GSA functions create three types of reports: Summary, Detailed and Both. The summary report lists the lowest transfer limits as calculated by the GSA function and the corresponding FCITC values. Ranking of limits from low to high are also indicated. The detailed summary report contains the worst generation dispatches with detailed flow reports for the limiting monitored element.

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Chapter 4
Transaction Impact Analysis

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4.1. Monitored Elements Impact Analysis

Transfer Limit Analysis > Impact Analysis > Monitored Impact Report

Figure 4.1. Monitored Elements Impact Analysis Dialog

The objective of the Monitored Elements Analysis is to provide a fast and simple set of functions to study the impact of transactions on specified monitored branches, interfaces, and flowgates. MUST provides two different approaches for monitored element analysis covering the most frequently asked question related to monitored elements response to various transactions.

The monitored element impact analysis allows the user to evaluate the impact of one study transfer on various monitored parameters. For example, it allows the user to evaluate flow changes due to scheduling additional transactions, report PTDF and OTDF factors for all monitored elements, identify monitored elements that can be impacted by the selected study transfer, and study OTDF factors variation for different contingencies.

The monitored element sensitivity analysis allows the user to evaluate the sensitivity of one or several monitored elements to various transactions. The study transfer participating points sensitivity report allows the user to evaluate the response of the selected monitored elements to all the participating points in importing or exporting subsystems. The transaction sensitivity report evaluates the impact of user defined transaction sets. The subsystem sensitivity report contains sensitivity factors for all specified subsystems. Monitored element sensitivity analysis provides a very efficient means to identify transactions which have the largest impact on the few selected or overloaded flowgates. This type of analysis provides the network sensitivity factors which will form the basis for Transmission Loading Relief (TLR).

4.1.1. Monitored Elements Impact Analysis

The Monitored Elements Impact Analysis is accessed by selecting Monitored Impact report under Impact Analysis group from the Transfer Limit Analysis Menu Item.

Three types of impact reports are available:

- Report for the base case (PTDF factors).
- Report for any user-selected contingency.
- OTDF sensitivity report allowing the user to study the variation of OTDF factors for the same monitored element under all contingencies.

Branch and interface impact reports are ordered by either:

- Bus numbers for branches or input order for interfaces.
- Monitored elements loading; to be exact, reports are ordered by the loading level at the maximum specified study transfer level.
- Absolute impact value.

The user can choose to eliminate reporting monitored elements loaded below a specified threshold or with the impact below a specified impact threshold.

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Contingency impact reports have a similar format to the base case structure report but contain more information allowing comparing flows and distribution factor for the base case and the selected contingency.

4.1.2. Monitored Elements Sensitivity Analysis

Transfer Limit Analysis > Impact Analysis > Monitored Sensitivity

Figure 4.2. Monitored Elements Sensitivity Analysis Dialog

The Monitored Element Sensitivity Analysis is accessed by selecting Monitored Sensitivity under Impact Analysis group from the Transfer Limit Analysis Menu Item. This feature allows you to specify several monitored elements for sensitivity analysis. You may choose a list of Branches, Interfaces or Flowgates by simply selecting the appropriate option button. Selecting each option activates the Select button from which the list of monitored elements may be selected. Monitored Elements Sensitivity dialog have a checkbox named "show old report". If this option is checked, the report will be in the text format. Otherwise, the reports will be in table format.

There are three different Monitored Elements Sensitivity Analysis reports:

- Subsystem Sensitivity Report contains sensitivity factors for all specified subsystems, which allows the user to evaluate the relative impact of the various subsystems on the selected monitored element. Subsystem sensitivity factors can be reported with respect to transfer to the specified reference subsystem or with respect to the system swing bus.

  The Subsystem Sensitivity Report is available in two parts: The Subsystems report and the Factors report. The "Subsystem" tab of the Monitored Sensitivity Dialog provides the report described above while the "Factors" tab provides specific data and sensitivity factors for a SINGLE subsystem. While the Subsystem Sensitivity Report provides side by side data for each subsystem specified, the Single Subsystem Sensitivity report provides detailed specific data for a single subsystem.

- Subsystems Participating Buses Sensitivity Report (see Figure 6-6) allows the user to evaluate the sensitivity of the selected monitored elements to all the participating buses in importing or exporting subsystems.

- Transaction Sensitivity Report (Figure 6-7) evaluates the impact of selected transactions. The transactions of interest are added or removed from the list by using the "Select" button just to the right of the "Only Selected Transaction" radio button. Alternatively, the "All Possible Transactions" radio button will compute transaction dispatch factors for all possible pairs of subsystems defined by the subsystem file. This report is convenient to quickly identify good and bad control area combinations and to filter out those transactions which have impact above a user designated cutoff.

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4.2. Impact Analysis on Areas, Zones, and Subsystems

Transfer Limit Analysis > Impact Analysis > Area/Zone/Subs Reports

Figure 4.3. Areas, Zones, and Subsystems Dialog

4.2.1. Applications of Impact Analysis

This group of functions provides reports on the incremental flow distribution caused by a user-specified study transfer. The functions also provide reports on initial and final interface flows before and after scheduling of a specified study transaction, a report on total systems balances, and a flow report for tie branches between selected areas, zones, and subsystems.

There are three levels of impact analysis:

- The Total Participation Factor (TPF) report, which summarizes the impact on areas, zones, or subsystems due to wheeling caused by the specified study transfer.
- The Interface Participation Factor (IPF) report, which summarizes the impact of the study transfer interfaces between areas, zones, and subsystems.
- The Ties report, which contains a detailed report on all tie branches between selected areas, zones, and subsystems.

The main objective of the impact analysis function is to evaluate interface flows between areas and zones. There are three alternative methods for specifying impacted systems: by areas, by zones, or by subsystems defined in the subsystem definition files.

Analysis by areas/zones is the most widely used. However, analysis by subsystems (e.g., based on subsystem definitions provided in the subsystem file) is more versatile and allows analysis for virtually any user-defined bus or set of buses. Therefore, the system description can range from global definitions such as regional systems (VACAR, MAPP, SPP, etc.) to detailed definitions such as portions of a utility. The impact of bus-to-bus transactions can also be analyzed.

One feature of the impact analysis is that it allows evaluation of interface flows between different types of subsystems (e.g., between areas and zones and even between overlapping systems). IPF for overlapping systems shows the percentage of flow that leaves the FROM system and arrives at the TO system.

By specifying a study transfer from one subsystem within a control area to another subsystem within the same control area, the effect of internal shifts in dispatch on the surrounding systems (loop flows) can be determined. By default, the area/zone impact analysis functions compute TPF/IPF/PTDF factors for the base case, but the user has the option to compute the same factors for any of the specified contingencies. The user may select a contingency by specifying the contingency number. A contingency ID number is reported with contingency labels on all FCITC reports and within contingency analysis reporting functions.

4.2.2. Invoking the Impact Analysis

The Impact Analysis on Areas, Zones, and Subsystems Analysis is accessed by selecting Area/ZonesSubs Reports under Impact Analysis group from the Transfer Limit Analysis Menu Item.

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The Area Impact dialog contains several information groups as described below:

- **Current Study Transfer Definition:** A study transfer must be defined before area impact calculations can be performed. If a study transfer has been defined earlier, the subsystems and the transfer level will appear as a default definition on the form. If a study transfer definition has not been defined, the program prompts the user to define one prior to computing an Area Impact report.

- **Type of Report:** The available options are: Total (TPF report, affects Impacted Systems only), Interface (IPF report affects both Impacted and Neighboring Systems), and Ties (Tie Branch report affects both Impacted and Neighboring Systems).

- **Impacted and Neighboring Systems:** Impacted systems are the areas, zones, or subsystems for which reports will be computed. Only impacted systems need to be specified for the TPF reports; the user does not need to specify Neighboring Systems.

Impacts on interfaces and tie branches are reported for interfaces between the selected Impacted Areas, Zones or Subsystems and the specified Neighboring Systems.

To select Impacted system(s), click on the appropriate option and then press the "Select" button to choose the specific system from the list of available systems. Similarly, you can choose the Neighboring system(s).

The Total report option does not use the Neighboring systems.

Optional parameters of the Area Impact Report are as follows:

- Sorting Method for Reporting TPF: Three options are allowed: by area/zone number, by the impact (e.g., TPF value), or by area/zone/subsystem name.

- TPF, IPF, and TIES Report Cutoff: Values to skip reporting subsystems and interfaces where the impact of the study transfer is less than the cutoff values.

- Report Impact for: Two reporting options are available: Base Case and Contingency #. If the user has selected to report a contingency case, the contingency number must also be specified. This number can be obtained from the list of user-defined contingencies and is also reported on various MUST reports.

Area Impact Report

The first column in this report shows the balance (i.e., the net interchange) of every reported area (MW). The second column contains TPF according to the NetArAr method. The third column contains TPF computed by the alternative AbsInOut method, which is always greater than or equal to the NetArAr TPF.

The TPF report can be sorted either by TPF values or according to area/zone/subsystem name or number. Impacted areas with a small TPF below the specified TPF cutoff will be ignored.

IPF Report

The IPF report shows the total flow changes on the interfaces between selected areas caused by a specified study transfer. IPF can be defined in two ways: as the arithmetic sum of flow changes on the ties comprising the interface, or as the sum of absolute flow changes. The first is used most often and expresses the change in interface flow between two areas due to the study transaction.

These two definitions provide different results only if the study transfer increases flow on some of the ties comprising the interface and reduces flows on others. The difference in values between two IPF definitions provides an estimate of incremental circulating flows within the interface. In the case where circulating flows

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within the interface is more than 0.1%, IPF reports contain additional messages on reversed flows and reports the total fraction of incremental flows opposite to the main direction of IPF change (RevSens).

The first column in this report shows initial flow between two areas/zones/systems. The second column shows the interface flow with the scheduled study transfer. The next two columns display IPF values for the net and absolute definitions. AbsIPF values are always positive, so the total sum by area is always greater than zero even for areas other than the source and sink areas.

Tie Branches Report

Branch level reports provide the most detailed information on PTDF/OTDF for every tie between selected systems. MUST automatically identifies all ties for selected interfaces and then reports the impact of the study transfer on all ties, including MW flows.

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Chapter 5
Contingency Analysis Functions

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5.1. Overview of Contingency Analysis Functions

The contingency analysis functions in MUST can be used to examine the effect of contingencies on the monitored elements. These functions include:

- DC Contingency Analysis
- AC Contingency Analysis

The DC contingency functions allow the contingency analysis of the pre-shift case for the specified contingency list as well as the contingency analysis of the study case after a specified shift, or after a group of transactions. The AC contingency functions can be used to carry out complete AC contingency analysis of the pre-shift case only.

The DC and AC contingency analysis functions are both broken-down into two principal areas:

- Analysis by Contingency
- Analysis by Monitored Element

Analysis by Contingency allows the listing of statistics and results for contingency analysis. Contingencies can be listed by various criteria, such as, the violations caused or the number of islands created by the contingency. Analysis by Monitored Element is used to generate monitored element lists and contingency analysis results, sorted by monitored element.

AC Contingency Analysis will require the user to execute a new contingency study or retrieve the results of a previously calculated study. AC Load Flow Solution options can be modified to adjust power system control action modeling and solution parameters such as iteration limits.

It is important in the execution of the contingency analysis functions that the user is aware of the MUST Options in effect during the calculation. For example, the choice of distribution factor cut-off and branch limit will obviously have a big impact on the results.

5.1.1. Contingency Severity Index

The Contingency Severity Index (CSI) is used to provide a numerical indication of the severity of each contingency. CSI index implemented in MUST provides a simplified solution satisfying the following criteria:

- If there are no post-contingent overloads and there is no loss of load/generation, voltage problems (in AC mode only), CSI is zero.
- A higher CSI indicates more severe overload or voltage problems (in AC mode only). A contingency with larger overloads or with several simultaneous overloads should have a larger CSI.
- If the initial load flow case has several overloads (with respect to contingent rating), then it is likely that most of the contingencies will have a non-zero CSI. The CSI can be used to find the worst contingencies (defined as contingencies with the highest CSI) and the best contingencies (reducing CSI, preferably to zero) that can be considered as corrective actions or operating guides.

In DC mode, Overload CSI provides the measure of the severity of a contingency based on postcontingency overloads and loss of load and generation:

$$CSI = ?_i(([P_i) - P_{lim_i}])^2) + LossOfLoad^2 + LossOfGener^2$$

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where:

P_i = Post contingent branch flow
$Plim_i$ = Contingent branch rating
LossOfLoad = Lost load
LossOfGen = Lost generation

It is very important to emphasize that summation in the above formula is applied to overloaded branches only, thus CSI will be zero if there are no overloads.

In AC mode, MUST reports two distinct CSIs for every contingency. Overload contingency severity index is very similar to the overload CSI for DC mode, but checks MVA rating violations for transformers and current overloads for transmission lines. Voltage contingency severity index is quite similar to overload SCI and is also applied for buses with voltage violations only.

$VoltCSI = ?_i((V_i - Vlim_i))^2)$

where:

$Vlim_i$ = Violated voltage limit (either high or low)

There are several reports by contingencies (both in DC and AC modes) that report CSIs for every contingency. MUST allows the user to sort these reports by severity index or to filter contingencies by severity indices above or below a user specified threshold.

In DC mode, several reports by contingencies show how the CSI changes after scheduling a defined study transfer to the specified test level.

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5.2. DC Contingency Analysis

Contingency Analysis > DC Contingency...

5.2.1. Analysis by Contingencies

Figure 5.1. Detailed Contingency Analysis - by Contingencies

DC contingency analysis (provides fast and complete screening of contingency violations using a DC load flow solution. It includes a summary/statistics analysis on specified contingencies and a parametric DC contingency analysis. This parametric analysis performs contingency analysis at zero and maximum specified study transfer levels simultaneously.

The user may:

- Create reports for the pre-shift (no transfers added) load-flow case
- Compare pre-shift DC contingency analysis results with the results when a study transfer is added to the pre-shift case, skipping the reporting of violations that are not impacted by the currently defined study transfer.

Several general functions are provided by the DC Contingency Analysis:

- Contingencies can be excluded/included at any time from further analysis by using the Exclusions / Select Contingencies to Exclude menu item. Using the Specific Contingency Option allows users to Exclude or UNExclude individual contingencies
- A contingency analysis summary report is available.
- A general statistics report providing summary information on the contingency analysis results can be generated.
- The user may define or forget a study transfer for the analysis using the Define/Undefine Study Transfer menu under DC Contingency analysis. MUST will provide results for conditions before and after the study transfer. The user may also remove or forget the study transfer.

To generate a DC Contingency Report, select Detail Contingency Analysis under DC Contingency from the Contingency Analysis Menu Item. The user may then filter the contingencies to be studied by selectitng an option under Select contingencies for analysis. You may select contingencies by:

- Category – Select a category, such as, contingencies With Violations or With Islands, from the drop-down list.
- Specific (All, N, N-N) – Enter the specific contingency number or range of contingency numbers.
- Above or Below Severity Index – Enter the severity index of interest.

In addition, you may enter Cutoff Values to filter out contingency results that are below your range of interest. These cutoff values may be selected from the MUST Options menu item.

Two cutoff values are available:

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- Contingency Case Flow Change Cutoff – This option is used for eliminating any limiting elements which are not significantly affected by a given contingency.
- Contingency Case OTDF Change Cutoff – This option was added for compatibility with PSS®E only. Users should be careful with this option as outages involving bus injection only (i.e., generator outages) do not change distribution factors at all.

The following reports are generated based on the option selected in the Contingency Options tab of the dialog.

- Selected Contingencies with Violations produces a report of violations for the current filtered contingency selection.
- Summary List of Selected Contingencies produces Contingency lists for the selected contingencies .
- Details for Selected Contingencies produces Detailed contingency analysis results based on the checked paramaeters. The user may specify the loading percent cut-off for this report. For example, if the user specifies 90%, MUST will report monitored elements with loading greater than 90% of rating. The user may also specify whether the events involved in each contingency as well as islanding and dispatch information should be listed in the report.

5.2.2. Statistics on All Contingencies

Contingency Analysis > DC Contingency> Statistics on All Contingencies ...

Statistic functions summarize statistics on all specified contingencies. This function classifies all contingencies in several categories.

5.2.3. Contingency Analysis Summary

Contingency Analysis > DC Contingency>Contingency Analysis Summary ...

This report can be used to compare the initial case and the final case with the study transfer at a specified value. Also, it can show violations that have response factors (OTDF) below a user-specified threshold. The MUST Options can be used to change the response factors threshold or to activate/deactivate the use of the response factor cutoff for violations analysis.

5.2.4. Selected Contingency with Violations

This function reports all contingencies with violations among a user selected subset of contingencies. Three reported columns contain contingency branch flow, total number of violations caused by the reported contingency, and the worst violation for the initial and final cases .

5.2.5. Summary List of Selected Contingencies

The Contingency Summary List Report provides summary information (one line per contingency) for user-selected contingencies. Also, this report provides contingency indices that can be used by other contingency analysis functions. Reviewing this report identifies potential data errors and provides a convenient overview on selected contingencies.

5.2.6. Details for Selected Contingencies

A Detailed Contingency Report provides the following information on selected contingencies:

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- Detailed contingency description including the listing of all contingency events. For branch opening events, it will report branch flow at initial and maximum study transfer levels. A user may suppress detailed reporting of dispatch events.
- List all the monitored elements with loading above a user-specified loading level threshold. Selecting 100% loading level reports overloads only.
- Island composition for contingencies that create buses which are not connected to the swing bus.

5.2.7. Analysis by Monitored Elements

Figure 5.2. Detailed Contingency Analysis - by Monitored Elements

A number of reports can be generated from Analysis by Monitored Elements. It also shows the number of the currently excluded contingencies according to the applied EXCLUDE files and contingencies.

Reports can be generated for branches, interfaces or flowgates. There are three types of reports available:

- The All Violations List option produces a list of violations. The user may specify the loading percent cut-off for this report. For example, if the user specifies 90%, MUST will report monitored elements with loading greater than 90% of rating. Also, the violations reports can be generated for all contingencies or for the reduced set of contingencies selected in the Analysis by Contingencies menu.
- The List Monitored Element option creates a list of monitored elements for the selected monitored element type.
- The Summary on Violations/Frequency option produces a summary report of violations.

As with the DC Contingency Analysis Dialog, several general functions are provided:

- Contingencies can be excluded/included at any time from further analysis by using the Exclusions / Select Contingencies to Exclude menu item. Using the Specific Contingency Option allows users to Exclude or UNExclude individual contingencies
- A contingency analysis summary report is available.
- A general statistics report providing summary information on the contingency analysis results can be generated.
- The user may define a study transfer for the analysis. MUST will provide results for conditions before and after the study transfer. The user may also remove or forget the study transfer.

To generate a DC Contingency Report, select Detail Contingency Analysis under DC Contingency from the Contingency Analysis Menu Item. The user may then filter the contingencies to be studied by selectitng an option under Select contingencies for analysis. You may select contingencies by:

- Category - Select a category, such as, contingencies With Violations or With Islands, from the drop-down list.
- Specific (All, N, N-N) - Enter the specific contingency number or range of contingency numbers.
- Above or Below Severity Index - Enter the severity index of interest.

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5.2.8. Summary of Violations/Frequency

In the case of a large number of violations under different contingencies, this report allows fast identification of multiple violations of the same constraints under different contingencies.

5.2.9. Violations Report

The Violations Report provides details for all monitored element flow violations. Preshift (and post-shift) flows are listed as well as branch ratings and the contingencies causing the overloads.

5.2.10. Monitored Element List Report

This function produces a list of all the monitored elements. The user may list all branches or all interfaces or all flowgates.

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5.3. AC Contingency Analysis

Contingency Analysis > AC Contingency...

A full AC contingency analysis of the post-contingency system is carried out for each contingency and the results are stored for reporting and review.

The MVA loading for monitored branches is calculated for pre- and post-contingency conditions and MUST reports on loadings greater than the MVA ratings selected in the MUST Options. Bus voltages can be monitored for high or low voltage levels and for large voltage magnitude changes as a result of contingencies. MUST will also indicate when voltage collapse and nonconvergence has been caused by contingencies. The user may select the AC load flow solution parameters for each AC contingency analysis study. The solution options include a Non-Divergent Load Flow Algorithm which offers insights into low-voltage and voltage collapse phenomena for the case being studied.

AC contingency analysis in MUST is performed on the pre-shift case and does not directly calculate the AC contingency results after a specified transfer or specified transaction(s).

The AC contingency analysis recognizes all contingency file formats used in DC contingency analysis, including contingency with branch closing, load throwover, various bus events, and dispatch commands. This analysis recognizes all monitored element file formats; including voltage deviations and changes, and flowgates.

5.3.1. Solve/Execute

AC Contingency Analysis is invoked by selecting Solve/Execute under AC Contingency from the Contingency Analysis Menu Item. MUST will solve for the AC conditions after each contingency in the contingency list. AC Contingency Analysis can be relatively slow. The progress of the AC Contingency Analysis calculation can be monitored by viewing the Progress window.

For each execution of the AC Analysis, the contingency flow change cut-off and contingency voltage change cut-off values can be selected and modified. These values are accessed through the AC Transfer & Contingency LF Options tab of the MUST Options dialog

In addition, the AC Transfer & Contingency LF Options provides access to other AC load flow solution options. You can select a Newton or Fast Decoupled solution, a non-divergent LF solution, control adjustments settings, along with selecting solution parameters. If a Fast Decoupled solution has failed, you may want to try the Full Newton method to attempt a convergence. The Non-Divergent Load Flow Algorithm may help solve otherwise unsolvable cases and may help minimize mismatches. Analysis of the non-convergent case allows the identification of low voltage problems.

Use of the the standard F10 interupt is supported. As the AC contingency analysis function may take a long time to execute, F10 interupt support serves as a mechanism to allow the user to terminate (i.e., cancel) the analysis if desired. AC contingency analysis function results will be lost when prematurely terminating the function.

5.3.2. Read AC Contingency Results File

As an Alternative to running the AC Contingency Analysis through Solve/Execute, you may enter a previously solved AC Contingency File by selecting the Read AC Contingency Results File menu item. Note, however, that this file will not read properly unless all of the current MUST Options, ACLF Options and Cutoff values match

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exactly the values used when the file was saved. Also, when reading back an AC contingency analysis file, the user must run the same load flow case and have defined the same monitored elements and contingencies.

Until an AC Analysis is either Executed or Read in, all of the AC Contingency Analysis reporting functions will be unavailable. Once a solved AC Analysis exists in memory, the reporting functions will become available.

Results of a successfully executed contingency analysis can be saved in a contingency results file using the Save ACCONT Results button. This file can be read back into MUST so that the contingency analysis results are available without re-executing the function.

5.3.3. Analysis by Contingencies

Figure 5.3. Detailed Contingency Analysis - by Contingencies

Several general functions are provided by the Analysis by Contingencies:

- Contingencies can be excluded/included at any time from further analysis by using the Exclusions / Select Contingencies to Exclude menu item. Using the Specific Contingency Option allows users to Exclude or UNExclude individual contingencies
- A contingency analysis summary report can be produced.
- A general statistics report providing summary information on the contingency analysis results can be generated.

To generate a AC Contingency Report, select Detail Contingency Analysis under AC Contingency from the Contingency Analysis Menu Item. The user may then filter the contingencies to be studied by selectitng an option under Select contingencies for analysis. You may select contingencies by:

- Category – Select a category, such as, contingencies With Violations or Excluded, from the drop-down list.
- Specific (All, N, N-N) - Enter the specific contingency number or range of contingency numbers.
- Above or Below Severity Index - Enter the severity index of interest.

As discussed previously under the Section "Contingency Severity Index", the Contingency Severity (CSI) index for overloads applies to the calculation for AC overloads with the introduction of a voltage CSI.

A number of reports can be generated from the Analysis by Contingencies menu:

- Selected Contingencies with Violations produces a report of violations for the current filtered contingency selection.
- AC Contingency Summary for Selected Contingencies will generate a summary for the selected contingencies.
- Summary List of Selected Contingencies produces Contingency lists for the selected contingencies .
- Details for Selected Contingencies produces Detailed contingency analysis results based on the checked paramaeters. The user may specify the loading percent cut-off for this report. For example, if the user specifies 90%, MUST will report monitored elements with loading greater than 90% of rating. The user may

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also specify whether the events involved in each contingency as well as islanding and dispatch information should be listed in the report.

5.3.4. Statistics on All Contingencies

Contingency Analysis > AC Contingency> Statistics on All Contingencies ...

Statistic functions summarize statistics on all specified contingencies. This function classifies all contingencies in several categories. This function also shows the number of the currently excluded contingencies.

5.3.5. Contingency Analysis Summary

Contingency Analysis > AC Contingency>Contingency Analysis Summary ...

This report provides summary information or branch overloads as well as bus voltage violations. The number of voltage violations is provided along with the magnitude and contingency number of the worst bus voltage violations.

5.3.6. Selected Contingency with Violations Report

This function reports all contingencies with violations among the user selected subset of contingencies. Three reported columns contain contingency branch flow, total number of violations caused by the reported contingency, and the worst violation for the initial and final cases.

5.3.7. AC Contingency Summary for Selected Contingencies

This report provides summary information (one line per contingency) for user selected contingencies. Also, this report provides contingency indices that can be used by other contingency analysis functions. Reviewing this report identifies potential data errors and provide a convenient overview on selected contingencies.

5.3.8. Details for Selected Contingencies

A Detailed Contingency report provides the following information on selected contingencies:

- Detailed contingency description including the listing of all contingency events. For branch opening events, it will report branch flow at initial and maximum study transfer levels. A user has the option to suppress detailed reporting of dispatch events.
- List all the monitored elements with loading above user-specified loading level threshold. Selecting 100% loading level will report overloads only.
- Island composition for contingencies that create buses, which are not connected to the swing bus.
- Bus voltage violations report allows the user to specify the loading percent cut-off. For example, if the user specifies 90%, MUST will report monitored elements with loading greater than 90% of rating. The low voltage and high voltage margins can be selected for the presentation of voltage results. The margin is the amount to be subtracted from the low voltage limit specified in the monitored element file (or added to the upper voltage limit/voltage change limit specified in the monitored element file). For example, if the monitored element file specifies that bus voltages below 0.95pu should be considered violations and the user enters a margin of 0.02 in the Vlow box, then MUST will report voltages less than 0.93pu. If the user enters zero margin, the bus voltage limits specified in the monitored element file apply for the report. The

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user may also specify a listing of events involved in each contingency as well as islanding and dispatch information.

5.3.9. Analysis by Monitored Elements

Figure 5.4. Detailed Contingency Analysis - by Monitored Elements

A number of reports can be generated from Analysis by Monitored Elements.

The formats of the AC Contingency Analysis Reports are very similar to the DC Contingency Analysis Reports. The AC reports are based on full MVA calculated flows. Also, the AC reports contain information on bus voltages. The following sections contain descriptions of the reports generated by AC Contingency Analysis.

As with the AC Contingency Analysis Dialog, several general functions are provided:

- Contingencies can be excluded/included at any time from further analysis by using the Exclusions / Select Contingencies to Exclude menu item. Using the Specific Contingency Option allows users to Exclude or UNExclude individual contingencies
- A contingency analysis summary report is available.
- A general statistics report providing summary information on the contingency analysis results can be generated.

To generate an AC Contingency Report, select Detail Contingency Analysis under AC Contingency from the Contingency Analysis Menu Item. The user may then filter the contingencies to be studied by selecting an option under Select contingencies for analysis. You may select contingencies by:

- Category - Select a category, such as, contingencies With Violations or Excluded, from the drop-down list.
- Specific (All, N, N-N) - Enter the specific contingency number or range of contingency numbers.
- Above or Below Severity Index - Enter the severity index of interest.

Reports can be generated for branches, interfaces, flowgates or Bus Voltages. There are three types of reports available:

- The Summary on Violations/Frequency option produces a summary report of violations.
- The All Violations List option produces a list of violations. You may specify the loading percent cut-off for this report. For example, if you specify 90%, MUST will report monitored elements with loading greater than 90% of rating. For voltage violation results, the voltage limit margins may be specified. The margin is the amount to be subtracted from the low voltage limit specified in the monitored element file (or added to the upper voltage limit/voltage change limit specified in the monitored element file). For example, if the monitored element file specifies that bus voltages below 0.95pu should be considered violations and the user enters a margin of 0.02 in the Vlow box, then MUST will report voltages less than 0.93pu. If the user enters zero margin, the bus voltage limits specified in the monitored element file apply for the report.
- The List Monitored Elements option creates a list of monitored elements for the selected monitored element type.

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5.3.10. Summary of Violations/Frequency

In the case of a large number of violations under different contingencies, this report allows fast identification of multiple violations of the same constraints under different contingencies. For example, if the same constraint is violated under all contingencies, it is likely a base case problem.

AC Contingency Analysis includes monitoring of bus voltages. Low voltage and high voltage violations are listed separately.

5.3.11. Violations Report

The Violations Report provides details for all monitored element flow violations. Listed are the branch ratings, the contingencies causing the overloads and the monitored bus voltage violations.

The voltage deviations listed in this report are defined as follows:

- Low Limit = Low Voltage Limit - Low Margin
- High Limit = High Voltage Limit + High Margin
- Voltage Drop Limit = Low Voltage Deviation Limit - Low Margin
- Voltage Rise Limit = High Voltage Deviation Limit + High Margin

where the voltage limits and voltage deviations are given in the Monitored Element File for each bus. The margins are entered in the Voltage Limits Margins text boxes on the AC Contingency Analysis Dialog from the Monitored Elements Options tab. On the report itself, the line just above the table details the margin adjustment values if any are non-zero.

Voltage limit violated values are displayed for both the base case and contingency case. Voltage deviation violated values are also displayed for both the base case and contingency case.

In addition, a test checks to see if the AC Bus Voltage violation limits are violated; i.e. the lower limit is equal or greater than the higher limit. These limits may become violated when the user adjusts the Voltage Limits Margins.

If these limits are violated, codes X,Y, or Z will be displayed in the column labeled “Viol”. A code of X indicates the voltage limits are in violation; i.e., Low Voltage Limit >= High Voltage Limit. A code of Y indicates the voltage deviation limits are in violation; i.e. Low Voltage Deviation Limit >= High Voltage Deviation Limit. A code of Z indicates that both the voltage limits and voltage deviation limits are violated.

5.3.12. Monitored Element List Report

This function produces a list of all the monitored elements. The user may list all branches, all interfaces, all flowgates or all bus voltages.

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Chapter 6
MUST Base Load Flow

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6.1. AC Load Flow Solutions

There are two load flow solution options available, optimized for use with MUST:

- A double-precision Newton-Raphson solution
- A Decoupled method

These two load flow methods have different solution characteristics but should result in basically the same solution, when converged. However, due to differences in solution and modeling of local controls (switched shunts, phase shifters, LTC transformers) methods may arrive at slightly different solutions.

Both of these functions are available from the MUST Base Load Flow Menu Item.

Non-Divergent Load Flow:

There are non-divergent options in all functions mentioned above (works best with LFNSOL) that may help solve some cases that cannot be solved by the PSS®E load flow.

Flat Start Solution:

Occasionally, load flow solution algorithms might have problems starting load flow solutions from a flat start. This is a common problem for large equivalent MMWG type load flow cases. The most common reasons for such problems are:

- Data errors
- Conflicting voltage objectives for electrically close buses.
- Large initial phase shifter angles (30 to 45°) or off-nominal TAP ratios of low-impedance transformers can cause divergence on the first iterations.

Double Precision:

LFNSOL and LFDSOL use double-precision arithmetic and can solve load flow to virtually any tolerance (i.e., 0.0001 MW). Often full Newton needs only 1-2 more iterations to reach such a tolerance. Most 12000 bus cases cannot be solved with single precision to tolerances below 1-5 MW mainly because of low impedance of some nonzero impedance lines.

LFNSOL and LFDSOL Limitations:

There are only two known limitations of the MUST load flow algorithms:

- MUST cannot handle area interchange control. To use area interchange control, the user should use PSS®E to resolve the load flow case with the area interchange control option turned on.
- LFNSOL and LFDSOL do not support direct TAP adjustment methods. Instead, LFNSOL and LFDSOL support a discrete TAP adjustment method.

6.1.1. OPTIONS

The OPTIONS within the MUST Base Load Flow Functions are set through the Intial Line Flows Solution Options tab of the MUST Options dialog.

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These options are normally set prior to solving the load flow and consist of four option groups for controlling the load flow algorithm, the representation of controls for various elements in the power system, and some additional advanced options controlling load flow solution. The default values corresponding to each of the options outlined below are either the current values of these options or assigned by MUST automatically.

The groups are:

ADJUSTMENTS	Change the control options for transformer taps, phase shifters, and switchable shunts.
GENERATOR	Change generator reactive limit control and the iteration number to begin var limiting for the load flow.
SOLUTION	Specify parameters relating to the algorithm including iteration limits, real and reactive power tolerance, whether or not to invoke the non-divergent load flow or use a flat start, reactive adjustment acceleration factor, and the number of decoupled iterations in the MIXED solution mode.

ADJUSTMENTS

This group provides for the following automatic control options during a load flow solution:

- Adjustment of transformer off-nominal turns ratio to hold the voltage within a specified band at a designated bus, or to hold reactive power flow through the transformer within a specified band.
- Adjustment of transformer phase shift angle to hold the real power flow through the phase shifter within a specified band.
- Adjustment of switchable shunt devices. When this adjustment option is disabled, discretely operating switched shunts are locked at their nearest step if necessary, and all switched shunts, including continuously operating devices, are held constant.
- Area Interchange Control There are three options for area interchange control: (1) “Disabled" - suppresses area interchange control (2) “Tie lines only” - area swings are adjusted with an area’s net interchange defined as the sum of the flows on all of its tie lines and (3) “Tie lines and loads” - area swings are adjusted where an area’s net interchange includes tie line flows as well as contributions from loads connected to area buses that are assigned to areas other than the bus’ area, and from loads assigned to the area that are connected to buses assigned to other areas.
- DC line tap adjustments

GENERATOR

This group allows the user to specify the handling of var limit controls for generator buses in a system. Initially, generator bus voltage will remain at the scheduled voltage as long as the var output at the generator bus does not violate the var limits. When a var limit is violated, the reactive power is set to the appropriate limit and the bus voltage will deviate as necessary from its scheduled value. If the bus voltage deviates from the scheduled voltage by more than a specified voltage control tolerance, actions will be taken to drive the voltage toward the scheduled voltage if this is possible without violating the var limits. MUST uses a value of 0.001 pu for the voltage control tolerance. This value is not accessible to the user.

the user can ignore the var limit at all type two buses except for those that have the same upper and lower limit values, by entering a VAR limit greater than the iteration limit. If the var limit check is enabled, the user can select the iteration at which the var limiting logic should be effective for either a flat start (all voltages

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set to 1.0 pu and angles at zero) or for a hot start (using the voltages in the working case as the initial estimate). Iteration number three is usually a good choice when starting from a flat start or resolving after a major change in the system. After a minor system change or when resolving to a finer tolerance, an iteration number of zero may be entered to make the var limiting logic effective immediately.

SOLUTION

This subfunction allows the user to set certain parameters relevant to the load flow solution algorithms. These parameters include:

Load Flow Iterations Number	Specifies the maximum number of load flow iterations to be executed during the solution.
Real and Reactive Power Tolerance	Determines the convergence of a load flow solution when no real or reactive component of bus mismatch exceeds the tolerance.
Reactive Adjustment Acceleration Factor	An acceleration factor is applied to the voltage adjustments made at generator buses. This is done to overcome the numerical stability problems that can arise when the Newton method encounters reactive power limits. The normal value of this acceleration factor is unity, but in cases exhibiting oscillatory convergence, it may occasionally be advantageous to decrease the acceleration factor. It will rarely be advantageous to set this acceleration factor greater than unity.
Low Voltage break point (pu) for load scaling	The constant power characteristic holds the load power constant as long as the bus voltage is above the defined voltage break point so that during the load flow solution when voltage dips below a user specified breakpoint, the load flow solution will assume an elliptical current-voltage characteristic of the corresponding load current for voltages below this threshold.
Max Iterations to Freeze Adjustments	Enabled controls will be locked after the number of iterations specified, which in some case helps to avoid oscillatory behavior, for example for large step switchable shunts with relatively narrow voltage band setting values at the controlled buses.

6.1.2. Non-Divergent Load Flow Algorithm

The occurrence of divergence in a load flow is manifested by extreme values of bus power mismatches and wide fluctuations in bus voltages. Consequently, no useful information can be extracted from the final state of the bus quantities to help identify if there are significant voltage or var problems in portions of the network.

To help prevent the load flow from diverging a straightforward algorithm is employed. During each iteration of the load flow solution, the bus power mismatches are computed and used to estimate the voltage correction (dV) desired. Voltages are then updated by the equation:

```
Vnew = Vold + mdV

       where:
Vold and Vnew = the voltages before and after the correction
m             = a multiplier between zero and one
```

In a conventional load flow algorithm, the value of m is kept at 1.0 throughout the solution. However, in the non-divergent load flow logic, m is decreased whenever the power mismatches are found to be increasing after a voltage correction. In other words, if applying the full estimated correction, dV, results in larger power mismatches, then the amount of voltage correction is reduced (by halving the value of m), new voltages are

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computed, and the new mismatches are checked again. This adjustment of m continues until the mismatches show a decrease, in which case the solution proceeds to the next iteration. If m has been reduced below a MUST tolerance without any sign of improvement in power mismatches, suggesting that the solution cannot be improved further, the load flow is terminated at that point.

If the load flow is halted by the non-divergent algorithm, the bus quantities, although not sufficiently accurate to provide a converged load flow solution, usually can give a relatively good indication of the state of the network. In particular, voltage collapse situations can be identified by localized areas of severe power mismatches and low voltages, with the remainder of the network left free of problems.

The quantity monitored between successive voltage corrections is the sum of squares of the real and reactive power mismatches for both full Newton and fast decoupled methods. The minor real and reactive mismatches associated with each reduction in m are displayed in the load flow convergence monitor.

The convergence characteristics of the Newton-Raphson iterative algorithm will change when using the Non-Divergent Load Flow Algorithm. In general, the cases that have converged solutions using the normal algorithm will also converge when the non-divergent feature is used. This is especially true for the full Newton method. On the other hand, the non-divergent feature may cause convergence in cases where the standard Newton load flow does not reach a solution. Also, since the automatic adjustments are performed independent of the major load flow iterations, the coexistence of the adjustments and the non-divergent algorithm may affect the convergence characteristics of this algorithm.

6.1.3. Full Newton Load Flow Algorithm

A fully coupled Newton-Raphson iterative algorithm can be used to solve for the bus voltage magnitudes and angles needed to satisfy the bus boundary conditions contained in the working case.

Prior to the start of load flow iterations, the algorithm first checks that each nontype four bus is connected to a type three (swing) bus through the in-service AC network. If any violations are detected, an appropriate message is printed and the algorithm is terminated. In the course of the solution, the full Newton algorithm handles the initialization, load modeling, generator reactive limit, and automatic adjustments according to user selection. At the completion of each iteration, the largest voltage change is checked against a "blow-up" tolerance, which has a default value of 5.0 pu. If the largest change exceeds this tolerance, an appropriate message is printed and this algorithm is terminated.

The convergence tolerance in the full Newton-Raphson solution applies to the largest bus mismatch each iteration. Convergence is assumed when no real or reactive component of bus mismatch exceeds the tolerance. The default convergence tolerance is 0.1 MW and Mvar, but a tolerance as large as 5 MW/Mvar may be sufficient for some purposes, such as in contingency analysis.

The following rules and convergence characteristics apply to the full Newton-Raphson:

- The full Newton-Raphson load flow will normally converge in a very few iterations on well-conditioned cases, achieving very small bus mismatches.
- Negative reactance branches are permitted in the network.
- The number of iterations required to reach the convergence tolerance is generally insensitive to system size.
- The solution may diverge if the initial voltage estimate is poor.
- Reactive power problems may cause poor convergence characteristics.

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- Enforcing generator reactive power limits too soon may cause the algorithm to diverge.
- The full Newton-Raphson method is intolerant of data errors and insoluble conditions in local areas of the network.
- When the Newton-Raphson method diverges, it fails catastrophically, giving no indication of where the problems are.

6.1.4. Fast Decoupled Load Flow Algorithm

The Fast Decoupled algorithm uses an iterative scheme in which the real power angle calculation is decoupled from the reactive power voltage adjustment. It is basically a Newton calculation in which each iteration consists of a pair of half iterations-first with the voltage magnitudes held constant and new voltage phase angles determined, then with the phase angles fixed and new voltage magnitudes calculated. The network admittance matrices are held constant throughout the solution (i.e., the Newton direction is calculated with a fixed slope, except for changes in bus voltage control statuses).

This algorithm handles the network connectivity checking option, initialization, load modeling, generator reactive limit, automatic adjustments, load, and the blow-up check, in the same way as function Newton.

The rules and characteristics governing the use of the Fast Decoupled algorithm are similar to those of Newton solution functions. In addition, the following apply:

- The decoupled algorithm is much less sensitive to a poor initial condition than the full Newton.
- The decoupled algorithm converges poorly if wide ranges of circuit X/R ratios exist in the network.
- When the network contains branches with resistance close to or greater than the reactance, the iteration may reach some mismatch level and then begin to diverge, usually slowly.
- As the mismatches are reduced, the rate of improvement on successive iterations is slowed.

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Chapter 7
Exclusion

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7.1. Overview Of Exclusion

Exlusion functionality is used for Transfer Analysis and Contingency Analysis. During the course of MUST studies, the analysis and display of FCITC sensitivity to transactions or generation dispatch changes assume that the most limiting contingency/monitored element pair is controlling. For example, all parallel transaction sensitivity charts are built using this assumption. Occasionally, a user may wish to disregard one or more of the most limiting contingency/monitored element pairs. This can be done via reading a exclude file or selecting excluded elements via exclusion selection dialogs. Exclusion can be used to simplify the redefinition of controlling contingency/monitored element pairs during a MUST session.

There are six functionalities to manage Exclusion:

- Selecting **Read Exclude File** forces PSS®MUST to read in the new exclude/change data file and apply the file adjustments. Several exclude data file can be read in one after another. For detailed explanation of exclude data file please see Section 2.5, EXCLUDE Data File Contents
- Selecting **Report All Exclusions** will report all combined exclusions for all previously specified exclude data files.
- **Save exclude File**, creates a new combined exclude data file. If the exclude data file contains the exclude initial violations specification or if a user excluded several contingencies and monitored elements using select exclusion elements dialogs , then the new combined exclude data file will contain the explicit definition of excluded monitored elements and contingencies. Saved combined exclude data file file can be reused by Reading Exlude File, user can store current exclusion list for later use. For detailed explanation of exclude data file please see Section 2.5, EXCLUDE Data File Contents
- Selecting the **Remove All Exclusions** removes all adjustments made by all previously read exclude/change data files. This function can be quite useful, if a user would like to apply a new exclude data file and ignore previous EXCLUDE specifications.
- User can specify and exclude elements by using **Select Elements to Exclude** dialog.

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7.2. Exclusion Of Elements

Exclusion > Select Elements to Exclude

Figure 7.1. Select Element to Exclude Dialog

This dialog allows the user to specify monitored and contingency elements to be excluded. The user may choose a list of Branches, Interfaces, Flowgates, AC Voltage Buses or Contingencies by simply selecting the appropriate exclude monitored element type via radio button. Different from other elements, AC Voltage Buses is activated if and only if Ac Execute has been run. Selecting each element type changes the list to display each of the selected elements. Additionally exclusion option of each element is represented in the rows. The list includes already excluded elements and added elements which are to be excluded. In each list, rows of already excluded elements represented with no color and rows of added excluded elements for exclusion represented with orange color.

- **Add Exclusion:** The user can add new exclusions by pressing the "Add Element" button. This brings up another dialog which displays all of the available elements according to selected element type in main exclusion dialog. This allows the user to add elements with the spesific exclusion options to the list by simply using the buttons in between the two list boxes to move items from the Unselected elements list to the Selected elements list. When the selections are completed, the user may either simply press "OK" to keep current choices or "Close" to return main exclusion dialog without any change. If exclude options are available, the user should choose one of the exclusion option before press OK. If the user press "OK" button selected elements will be added to the list with the selected exclusion option. Figure 8.1 illustrates this selection Dialog. This illustration displays a list of Interfaces because the "Interfaces" option button was selected. By using filter button, unselected element list can be updated with elements that include filter text. This list would be replaced with Branches or Flowgates if one of the other options were selected.
- **Remove Exclusion:** The user can remove exclusion from the list by clicking "Remove Exclusion". Before clicking Remove Exclusion user should select exclusion to be removed. Selected exclusion will be represented with blue color. The user can remove only one exclusion for each time.

Figure 7.2. Figure 8.1 Selection Dialog of Exclusion Element

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Chapter 8
Advanced SCED/SCBD Applications

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8.1. Introduction of Advanced SCED/SCBD

Advanced SCED/SCBD is different from Conventional application of SCED and Conventional contingency analysis.

1. Conventional application of SCED is to minimize generation cost and fix branch overloads, but which is only focusing on the ***base case*** study; Advanced SCED/SCBD is focusing on ***contingency cases*** study.

2. Conventional contingency analysis is focusing on ***assessment*** that is to identify if the contingency case will create overloads; Advanced SCED/SCBD, besides to identify overloads, is focusing on the ***solution*** that how to fix (remove) overloads for a large group of contingencies.

8.1.1. Two Types of Solutions

Advanced SCED/SCBD has to two types of solutions.

1. If fixing overloads only for a single contingency case the solution is referred as *corrective case* calculation;

2. If fixing overloads simultaneously for a large group of contingency cases the solution is referred as *preventive case* calculation.

8.1.2. Applications

The two solutions can be applied for the contingency control in real-time operations.

- The *preventive case*can be applied for pre-contingency control, which is to pre-redispatch required generations that can prevent any overloads produced by a large group of contingencies.

- The *corrective case* can be applied for post-contingency control. If a overload is only able to be fixed by load shedding, it should be resolved in post-contingency control. After the corresponding contingency case occurs the *corrective case* can tell the operator whether and how much a load shedding should take place.

Besides the*preventive case* can also be used for very fast calculation of n-1-1 contingency analysis.

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8.2. Advantages of Advanced SCED/SCBD

8.2.1. Very High Performance

It is well known it's a big challenge in performance for non-linear SCOPF to calculate the preventive case on a large group of contingencies in a very large network. For example, ISO's' cases usually have over 80,000 buses and 100,000 branches, if the number of contingencies is larger than several thousands, it may take hours by non-linear SCOPF. However, the time can reduce to several seconds by Advanced SCED/SCBD.

The reason is very simple. If a line or several lines out of the service in a very large network, more than 99% lines would have zero influence and the corresponding calculations become unnecessary. Therefore, if we have a right algorithm that can successfully filter out of all unnecessarily calculated lines, as well as accurately calculate changes in rest of lines, the performance would be significantly improved.

8.2.2. Accurate Results

Tests show the results of Advanced Contingency Analysis achieves are nearly identical to that produced by AC Loadflow with the difference of only 2%~3%.

How to achieve excellent speed without sacrificing accuracy? It is by three components in the applications

1. Hybrid AC & DC branch flow model
2. Advanced Algorithm
3. Comprehensive search process

The details will be presented in following section.

8.2.3. Divergence Free in Calculations

During contingency calculations, very large overloads can produce divergence in AC power flow or non-linear SCOPF, which would result in corresponding solutions never found, regardless if which exist in the real world.

In Advanced SCED/SCBD only the base case is calculated by AC power flow, as long as the solution exist that always can be identified.

8.2.4. Voltage Violation.

Large overloads can create the significant voltage violations in contingency cases. As branch flows in contingency cases are calculated by hybrid AC & DC model, it is unable to calculate voltage violations.

However Advanced SCED/SCBD is focusing on the solution not contingency state, after overloads are removed in the solution the corresponding voltage deviation from the base case would also come back to the value of the base case with a very limited deviation.

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8.3. Three Major Components

Advanced SCED/SCBD uses Dual Revised Simplex with enhanced algorithms and strategy to calculate branch flow, and sensitivity vector *S* that is used in Simplex to correct violation.

8.3.1. Hybrid AC & DC Branch Flow Model

Branch flow = Base Flow + $\Delta Flow_{contingency}$ + $\Delta Flow_{correction}$.

1. Base Flow is the branch flow in the base case
2. $\Delta Flow_{contingency}$ is the flow change caused by contingency events
3. $\Delta Flow_{correction}$ is the flow change caused by generator re-dispatching or load shedding

It means each branch flow has three components.

- Base Flow is calculated by non-linear algorithm (AC Loadflow) which is accurate and never changes in all contingencies
- Two $\Delta Flow$ are calculated by linear algorithm
- Each contingency case only needs to calculate two $\Delta Flow$ and in a very large case more than 99% of them have almost zero values.

So the Hybrid Model guarantees more than 99% of branch flows are accurate that are produced by AC Loadflow; it is also convenient to filter out of all branch flows which is unnecessary to be recalculated in contingency cases.

8.3.2. Advanced Algorithms

Improve Accuracy of two ΔFlow

Enhanced Distribution Factor is applied to calculate$\Delta Flow_{contingency}$ and $\Delta Flow_{correction}$

1. The conventional LODF is calculated by the compensation method, ignoring interactions among multiple contingency events that could result in large inaccuracy, especially if multiple contingency events are close each other in the network. ENDF for $\Delta Flow_{contingency}$ is calculated by Sherman–Morrison formula without any approximation:

$$(A+uu^T)^{-1}=\left(I-\frac{uu^T}{1+u^Tu}\right)A^{-1}=A^{-1}-\frac{uu^T}{1+u^Tu}A^{-1} \quad (8.1)$$

   where $A \in R^{nxn}$ the original Matrix for DC model, $u \in R^{nxk}$is the modifying matrix.

2. Enhanced Distribution Factor for $\Delta Flow_{correction}$ is calculated by differential and partial differential of the matrix, which takes account of interactions between contingency events and correct actions.

Improve accuracy of sensitivity vector S

Sensitivity vector *S* is of sensibility factors regard to MW changes in each re-dispatched generator or shredded load to MW change in the violated branch. It is used in Dual Revised Simplex for overload removal. The larger the factor is, the more effective the overload can be corrected.

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To remove overload for the base case each sensibility factor only takes care of the sensibility of the corresponding generator/load to the overload. However, if it is to remove overloads for a contingency case, each sensitivity factor should also take account of interaction of the corresponding generator/load to contingency events, especially if a generator/load location is near to a contingency event the sensitivity could be 100% larger than that in the base case.

Differential and partial differential of the matrix are also applied to calculate the sensitivity vector.

8.3.3. Comprehensive Search Process

Diagram of comprehensive search process

1. The Preventive Case calculation use a Comprehensive Search Process to identify the largest overload in all contingencies.
2. By fixing the larger overloads all the smaller overloads can be automatically fixed and thus significantly reduce calculations required
3. In each event, only branches with nonzero `ΔFlow` need to be checked
4. A Preventive Case that may have taken hours to calculate by non-linear SCOPF, would be reduced to seconds by advanced contingency analysis.

8.3.4. Exception

Tests show results produced by Advanced SCED/SCBD are nearly identical to that produced by AC Loadflow, the difference is about 2%~3%.

The only exception is if a flow is heavily overloaded by Mvar, which usually happens on branches in power plants. As shown in the following diagram the voltage at the bus 344850 is remotely regulated by five units, if one of two transformers out of service, another would be heavily loaded by Mvar, regardless very large MW reductions carry on in units.

However, the test also further shows if the voltage regulation by units are changed to local buses, the large Mvar will be removed and Advanced SCED/SCBD can accurately correct the overload.

In such case, the identified units will be flagged with "!" in results

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8.4. How to Run Advanced SCED/SCBD

8.4.1. Options Window

There are six sections on advanced SCED/SCBD options window

1. Chose to run SCED or SCBD (no difference if generator data is not available).
2. Study Option
   - Run Corrective Case (for Summary)
   - Run Preventive Case
   - Run Load Shedding
3. Four Controls to remove overloads
   - Generator
   - Load
   - Phase Shifter
   - DC Line
4. Basic Options
   - Run Merit Order (not check if generator data is not available)
   - Run Security Constraints (to remove overloads)
   - Normal Branch Rating
   - Emergency Branch Rating
5. Additional Options
   - Total Area Generation Persistence (to maintain the total generation)
   - Security Margin (additional margin to reach the limit with maximum 0.15)
6. Input Files for generator economic data (if Run Merit Order is selected, otherwise leave empty)
7. Output files (very slow)
   - Base Case (after Run Merit Order)
   - Preventive Case

8.4.2. Data Format of Merit Order

The module is using a file format similar to the PSSE ECDI file with a little modification.

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Each machine to be dispatched by activity SCED/SCBD must have a data record specified for it

in the Economic or Block Dispatch Data File in the following format:

I,ID,PRIOR,FUELCO,PMAX,PMIN,HEMIN,X1,Y1,X2,Y2,X3,Y3,X4,Y4,X5,Y5,X6,Y6

Table 8.1. where

I	Bus number; bus I must be present in the working case with one or more machines connected to it.
ID	One- or two-character machine identifier of the machine at bus I for which data is specified by this record. ID = 1 by default.
Block	Set a number greater than zero if applying block dispatch; Machines with a lower Block# has higher priority for dispatch generation. PRIOR = 1 by default.
FUELCO	Fuel cost for the machine in dollars per MBtu if applying economic dispatch. No default is allowed.
PMAX	Maximum machine active power output; entered in MW. If this machine is the principal unit of a dispatch group, this is the sum of the maximum outputs of all machines in the dispatch group at the bus. If PMAX is not specified, PMAX and PMIN for this machine are set to the power limits contained in the working case (refer to PSSE Generator Data).
PMIN	Minimum machine active power output; entered in MW. If this machine is the principal unit of a dispatch group, this is the sum of the minimum outputs of all machines in the dispatch group. If PMAX is not specified, the value specified for PMIN is ignored and the one from the working case is used; otherwise, no default is allowed.
HEMIN	Minimum heat input required by the machine when in-service; entered in MBtu/hr. If X1 (see below) is greater than zero, HEMIN should be specified as the heat input required at X1 MW. No default is allowed.
Xi, Yi	Points on the incremental heat rate curve; Xi values are entered in MW and Yi values are entered in Btu/kWh. At least two points, and up to six points, may be entered. Both X and Y must be in ascending order, with X1 < PMIN and Xn > PMAX. If this machine is the principal unit of a dispatch group, this curve is the combined curve of all machines in the dispatch group.

8.4.3. Corrective Case Results

Corrective Case Tittle:

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Case Solution for Each Case:

Case Summary:

8.4.4. Preventive Case Results

Case Tittle:

Calculations:

Solution and Summary:

8.4.5. Load Shedding Results

Case Tittle:

Load Shedding for Each Case:

Case Summary:

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Chapter 9
MUST Excel Add-in

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9.1. MUST Add-in for Microsoft Excel®

The MUST Add-in for Excel is provided to assist users in transferring MUST reports from PSS®E to Microsoft Excel for post-processing. It can be used to transfer reports from earlier versions of MUST as well as current versions. It will import reports generated in the traditional .lis format familiar to users of earlier versions of MUST. It will not import the latest PSS®E Tabular reports as those reports are aleady formatted for Excel and need only be copy/pasted into Excel without the need for special processing.

To use the MUST Add-in, you need to first install it into your own version of Excel and then invoke it from the function tabs shown in the top row of your Excel Application. To install the MUST Add-in follow these steps:

- Open Excel and select File then Options.
- Select "Add-ins" from the left panel then near the bottom from the "Manage" drop down box, select "Excel Add-ins".
- Press the button labeled "Go ...". This will bring up the Excel Add-ins Dialog.
- Select "Browse" from the right side of the dialog and navigate to the PSS®E PSSBIN directory to locate the MUST Add-in.
- Choosing the MUST Add-in will add it to the list of available Add-ins in Excel and will be listed with the check box checked. Press OK.

You will now see a PSS®MUST tab at the top of your Excel Application. The Add-in is now ready for use.

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9.2. MUST Report Importer

Reports generated by the MUST Engine may be imported into Microsoft Excel using the PSS®E MUST Excel Add-in. MUST Reports generated from PSS®E need to be written to a file before importing. This is typically done by redirecting PSS®E output or from running a case using a script. When done in this manner, the PSS®MUST Report Importer may then be used to import these previously saved report files into Excel.

The Import Reports Feature is invoked from the "Import Reports" button of the PSS®MUST Excel Add-in. Select "Import Reports" to display the "Import MUST Reports" Dialog. From this dialog, you may select a computer drive and directory path to search for importable reports. Pressing the button with the ellipsis (...) will produce a drop down box from which you may select a valid drive or directory. The Up Button (^) allows you to select a directory above the one you are currently looking at. As you visit various directories, the report list will be populated with valid MUST reports found in that particular directory. Note that ONLY THOSE REPORTS RECOGNIZED BY MUST AS IMPORTABLE WILL BE DISPLAYED IN THIS LIST even though there may be other files in this directory.

Once you are in the desired directory, simply select the reports you wish to import from the list and select "OK". The "Select All" button allows you to select all of the reports in the list. Each report selected will be generated directly through the MUST Excel Interface in MUST format. The report Sheetname in the Excel Workbook will be preceded with the underscore (_) to distinguish it as an imported report.

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9.3. Contingency Analysis Report Comparison Tool

Each Contingency Analysis Report provides description and result parameters which identify the impact of contingencies on the system elements. Each report, however, provides data relevant only to the particular conditions of that specified run. Often many runs are required in order to determine the configuration with the minimum/maximum impact. The Contingency Analysis Report Comparison Tool is used to pull out key parameters from the reports of each of these runs and provide them in a summary table to allow a side-by-side comparison of parameters of interest.

This comparison tool can be accessed from the "Contingency Comparison" button of the PSS®MUST Excel Add-in. Before performing the actual comparison summary, however, you will need to separately generate each report to be compared and rename each respective worksheet to create a set of unique comparable reports.

For example, say you wish to compare several contingency Violation Reports. You would first, of course, read a case into PSS®E and MUST. Once the case is set up, you would select "Contingency Analysis" tab from the MUST drop down menu, choose DC contingency or AC contingency as appropriate and select Detail Contingency Analysis followed by "Selected Contingencies With Violations". This will generate a report with the tab "Cont Viol Sum". If you change your case and simply repeat these steps, you will generate another report with this new case. If you redirect each of these reports to a separate file you can import them into Excel using the Report Importer as described in the previous section. Once imported into Excel, these reports are avaliable for the Report Comparison tool to utilize.

Once you have generated and imported a collection of comparable reports, select the "Contingency Comparison" tab of the PSS®MUST Excel Add-in. This will bring up a dialog from which you may specify which worksheet should be the Master Report (i.e. Case 1), which worksheets to select to compare against the Master (and each other) and which columns of data to compare. Depending on the size of your report, it may take a while to collate all of the data for comparison. Selecting specific columns of interest will speed up the process by limiting the amount of collation required.

The resulting comparison summary will identify each report by its header data as Case 1, Case 2, etc. The header data is all of the information listed above the gray columnar bar common to each report. The comparison summary will then list each element to be compared and the comparable data from the column(s) you selected in the Compare Reports Dialog. The elements listed are drawn from the "Primary Column" shown in the Compare Reports Dialog. The "Primary Column" is derived from the type of report and is not user selectable.

On all Violations Reports, the Contingency Report Comparator will sort out the Contingencies related to each particular Monitored Element along with sorting the "Primary Column". This allows a more direct comparison between cases involving contingency violations, in addition to sorting out and comparing data for monitored elements.

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